Work2Skill: Capturing Human GUI Experience for Reusable Agent Skill Induction

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Paper version: arXiv 2610.12984v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *unfamiliar software and workflows requiring human domain knowledge remain challenging*

Human demos speak the GUI's language, but reuse needs grounded actions and rules split from task specifics. Work2Skill pairs a recorder, a dataset and an induction agent to close the gap.

Why reuse is hard

1. **Ambiguous actions.** Coordinates say where, not what the control is.
2. **Unfair comparisons.** Prior studies differ in demos, tasks, protocols.
3. **Fixed pipelines.** Preset order; recordings overflow context.

Evaluation setup

MyPCBench Bounded · 64 | OSWorld-Verified · 286 | GPT-5.5 · Opus-4.8 | Qwen3.5-35B · 9B | 100-step budget | 350 demos · 8.48 h

- **All online:** VM execution, official evaluators.
- **Same demos, same settings** per benchmark; only the guidance text differs.

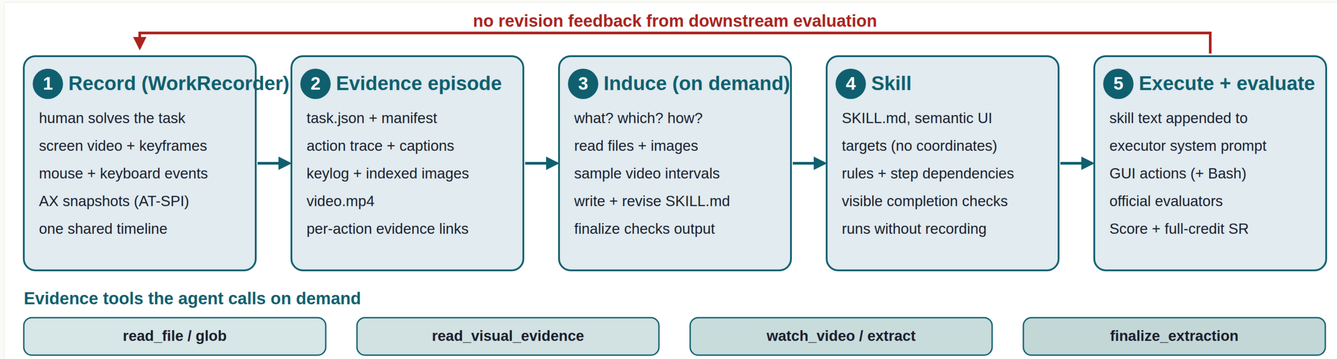
Our read (CUA Radar)

64	★★	online
score of 100	band	evidence level

- **Reports:** online eval, adapted strong baselines, induction-token cost, evidence ablation, code. **Not reported:** seeds/variance, budget match.
- **Flags (AP5):** demos collected on the eval tasks; RQ1 scores those same tasks.
- Sub-pattern R03. Authors' numbers, not rerun.

Method: record once, induce on demand, execute from text

- **Record:** video, keyframes, mouse/keyboard events, AX snapshots on one timeline (WorkRecorder).
- **Induce:** NexAU agent reads traces, images and video on demand; writes and revises SKILL.md.
- **Execute:** skill text appended to the executor prompt; GUI actions (+ Bash on MyPCBench).



Our redraw (not the paper's figure): record → evidence episode → on-demand induction → text-only skill → online execution.

$$Score = \frac{100}{|D|}\sum_i g_i \qquad SR = \frac{100}{|D|}\sum_i \mathbf{1}[g_i = 1]$$

Eq. 1: online Score and full-credit SR

Key results: online score

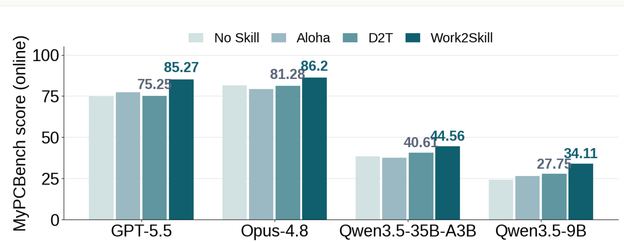


Table 2, redrawn: mean score (0–100), 64 MyPCBench Bounded Action tasks, official evaluator, per executor. Highest Score and SR of all methods, all four executors.

vs No Skill: +4.67–10.54 (MyPCBench), +3.11–7.50 (OSWorld-286)

Reuse: Jitter variants (N = 64)

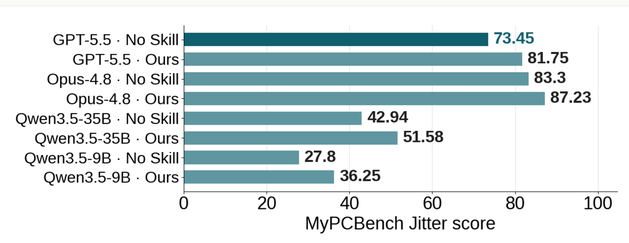


Table 3, redrawn: paired variants change entities, parameters or constraints in the same apps; skills reused unchanged.

SO WHAT → *Skills outlive their instance: +3.93–8.64 score and +6.25–9.38 pp SR vs No Skill on all four executors; steps drop by 2.49–5.44. The advantage over the stronger demo baseline is 4.28–6.25; SR ties D2T on Qwen3.5-9B.*

Headline Numbers

85.27
MyPCBench score, GPT-5.5

+4.67–10.54 vs No Skill
+3.11–7.50 OSWorld-286
350 demos

The paper's own controls

CONTROL	RESULT	READING
Evidence ablation	score −5.02–10.63	SR −14.06–21.88
Pipeline prompt	+3.97 score (ours)	steps 48.98→47.38
OSWorld · GPT-5.5	81.97 / 78.67 SR	D2T 80.32 · NS 74.47
Cross-model	GPT-5.5 inducer	lead up to 9.46

Takeaway

Record once; on-demand induction turns human GUI work into text skills that lift every executor.

reused on Jitter tasks | transfers across models | text-only, no training | code + data released

Scan to Read

Paper | Code